
An Interpretable Memory Decision Controller for LLM Agents Based on Three-Signal Complementarity: Decoupling Confidence and Consistency

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Abstract

Memory systems for large language models have focused predominantly on efficient retrieval, whereas the decision of whether retrieved memories should be trusted has received comparatively little attention. When the memory store contains conflicting positions, standard retrieval-augmented generation (RAG) blindly injects memories and amplifies hallucinations: in models susceptible to memory injection, the RAG hallucination rate under conflicting memories is markedly higher than that of a memory-free baseline. Inspired by memory signaling mechanisms in the prefrontal cortex, we propose the Memory Decision Layer (MDL), a zero-parameter memory decision controller situated between the retrieval and generation stages. Its core is a three-signal complementary encoder that fuses relevance, reliability, and task risk through QR-based orthogonal subspace projection and a meta-working-memory signal into an interpretable decision representation that quantifies the trustworthiness of retrieved memories. Building on this encoder, MDL explicitly decouples confidence from consistency and introduces risk inversion and explicit abstention. Evaluations on mainstream large language models and multiple open-source datasets show that MDL reduces the hallucination rate under conflicting memories by about 56.04% in general scenarios and approaches zero hallucination in high-risk scenarios. The controller is fully white-box: it relies purely on geometric operations, requires no trained parameters, and adds only about 0.14 ms per decision—roughly $50\times$ faster than the embedding-retrieval step that precedes it and four to five orders of magnitude faster than an LLM self-evaluation call.

1 Introduction

Large language model (LLM) agents rely heavily on external memory systems in multi-turn interaction and complex task solving. Retrieval-augmented generation (RAG) has become the dominant memory paradigm: knowledge is stored in an external corpus, semantically similar memories are retrieved, and they are injected into the generation context. Deciding correctly whether the retrieved memories can be trusted is a key prerequisite for the reliability of agent outputs.

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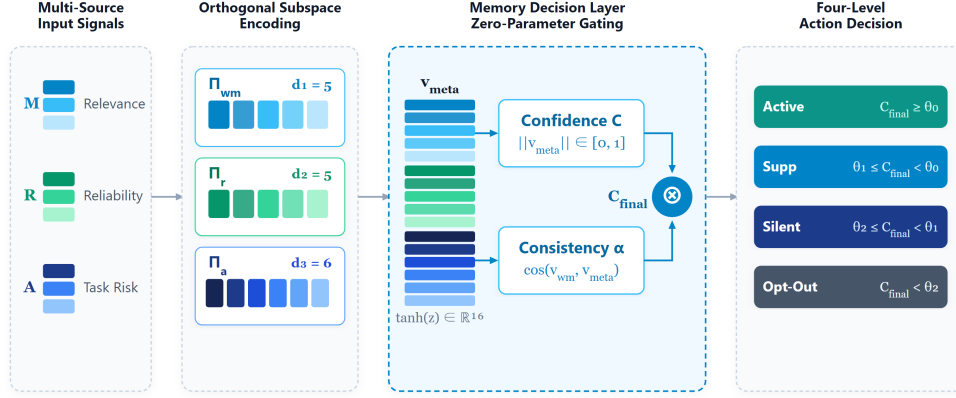


Figure 1: **Complete four-stage architecture of the Memory Decision Layer (MDL):** 1 multi-source input signals, 2 orthogonal subspace encoding, 3 the memory decision layer, and 4 four-level action decision.

Retrieving a highly relevant memory, however, does not imply that the memory is trustworthy. An outdated pharmacopoeia is adopted because of its high relevance, yet its advice is obsolete and dangerous; deprecated API documentation is emitted because it appears so frequently; New York case law is mistakenly cited in a California legal consultation. These cases share a common signature: high retrieval confidence but low substantive consistency between the memory and the task direction—the retriever only answers *which* memory is most relevant, but never whether that memory is trustworthy.

Worse still, when the memory store contains conflicting positions, standard RAG amplifies rather than mitigates hallucinations, a failure mode systematically documented in the knowledge-conflict literature [31, 32]. In end-to-end experiments on the TruthfulQA dataset [16], when the memory store contains both correct answers and common misconceptions, the hallucination rate of RAG reaches 53.0%, significantly higher than the 23.0% of the memory-free baseline ($p = 0.007$). The root cause is *retrieval as adoption*: the system unconditionally injects every retrieved memory without any trust assessment, and the model is misled by erroneous memories in roughly half of the cases. This quantitative gap indicates that the post-retrieval memory decision stage is an indispensable component of any trustworthy memory system.

Most existing studies focus on retrieving more efficiently or on repairing outputs after generation, leaving the trust-decision stage between retrieval and generation largely vacant. GraphRAG [10], MemGPT [2], and Mem0 [1] optimize storage and retrieval; HippoRAG [19] organizes human-like long-term memory in a graph structure. Self-assessment approaches such as Self-RAG [9] and Reflexion [5] defer trust verification until after generation, by which time erroneous memories have already contaminated the context and can only be remedied, not prevented; Huang *et al.* [28] further show that LLMs can hardly self-correct reasoning errors without external feedback. Post-generation verification alone therefore cannot fundamentally stop erroneous memories from being injected.

Filling this gap requires judging whether to adopt or abstain—before memory injection and using only retrieval-stage signals. This requirement maps naturally onto a recent neuroscience finding: Ning *et al.* [17] observed in the macaque prefrontal cortex (PFC) that metacognitive memory judgments are not driven by a single memory-strength signal; instead, neural populations encode complementary components—working-memory strength, trial history, and arousal level—in nearly independent functional subspaces and fuse them into a meta-working-memory signal that guides behavior and abstention decisions. This computational motif—assessing trustworthiness before consuming information—provides the legitimacy for inserting a memory decision layer between retrieval and generation.

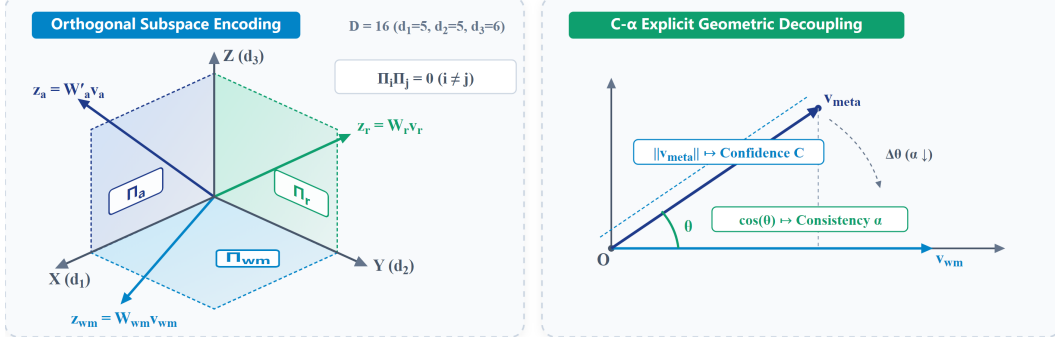


Figure 2: Illustration of the four-stage gated decision and action-mapping mechanism.

Inspired by this finding, we propose the Memory Decision Layer (MDL) (architecture shown in Fig. 1): it sits between the retrieval and generation layers and answers, with zero trained parameters, whether retrieved memories are trustworthy and whether they should be adopted. The core of MDL is a **three-signal complementary encoder**: it combines three complementary signal channels—relevance (M), reliability (R), and task risk (A)—projects them onto orthogonal subspaces, and fuses them with a meta-working-memory signal into a unified interpretable decision representation $\mathbf{v}_{\text{meta}}$. The risk signal A is a task-level engineering extension that captures the high-stakes consequences of domains such as healthcare, law, and finance. On top of this encoder, MDL further exploits the explicit decoupling of confidence C (norm) and consistency α (cosine), value encoding, and risk inversion ($s_A^{\text{inv}} = 1 - A$) to strengthen auditability and safety redundancy—among these properties, the C - α decoupling is the most prominent. MDL proceeds in four progressive stages: (1) signal extraction and risk inversion; (2) orthogonal subspace projection; (3) meta-working-memory fusion with C - α decoupling; and (4) four-level action mapping. Purely geometric operations guarantee ultra-low latency, and the explicit scalars C and α provide an interpretable attribution-and-audit interface.

The main contributions of this article are as follows:

- (1) **Three-signal complementary encoder**: We propose a zero-parameter memory decision encoder that integrates relevance (M), reliability (R), and task risk (A), placed between the retrieval and generation layers; orthogonal subspace coding and meta-working-memory fusion jointly characterize the trustworthiness of retrieved memories. An information-theoretic decomposition shows that fusing the three signals yields an observable synergy gain, and that the M signal provides irreplaceable safety-abstention value in the low-relevance regime.
- (2) **C - α decoupling mechanism**: On top of the encoder, we design orthogonal subspace projection and a gating mechanism that expose confidence C (norm) and directional consistency α (cosine) as two independently auditable scalars. Combined with risk-inversion encoding ($s_A^{\text{inv}} = 1 - A$), high-risk or conflicting memories spontaneously trigger abstention.
- (3) **Zero-training, ultra-low-latency deployment**: The controller relies entirely on geometric operations (QR decomposition, vector norms, and cosine angles), requires no trained parameters, achieves a single-decision latency of 40.1 μs (about 0.14 ms including lexical signal aggregation), and outputs the C and α scalars as an explicit attribution-audit interface.
- (4) **Cross-model end-to-end hallucination suppression**: We conduct extensive validation on TruthfulQA and HaluEval. MDL reduces the hallucination rate under conflicting memories from 53.0% (RAG) to 23.3% ($p = 0.014$) in general scenarios and achieves near-zero high-risk hallucination rates across mainstream LLMs including gemma-4-E4B-it, deepseek-v4-flash, and gemini-3-flash-preview.

2 Related Work

This section classifies related work along a single axis—the stage at which trust decisions about retrieved memories are made—into four categories, and closes each category with a precise statement of the difference between it and our work. Table 1 contrasts the proposed MDL with representative methods along capability dimensions.

2.1 Memory Organization and Retrieval Optimization Before Generation

These systems optimize memory storage and retrieval before generation, but make no decision on the trustworthiness of already-retrieved memories. Standard RAG drives generation with top- k semantic recall; GraphRAG [10] augments retrieval with a graph structure; HippoRAG [19] and its extension [20] organize knowledge graphs with personalized PageRank into human-like long-term memory; MemGPT [2] manages the context through paged main/external memory; MemoryBank [6], Generative Agents [3], Larimar [22], ExpeL [23], Mem0 [1], A-MEM [21], and LongMem [7] extend memory systems from the perspectives of forgetting curves, memory streams, knowledge editing, experience distillation, agentic memory organization, and long-term capacity, respectively; Adaptive-RAG [30] adapts the retrieval strategy to question complexity. The shared limitation of these systems is that the decision relies entirely on a single retrieval score: when the retrieval score is high but directionally wrong, the memory is still adopted unconditionally.

2.2 In-Generation and Post-Generation Self-Assessment

These methods self-assess outputs during or after generation, but by then erroneous memories have already been injected into the context; they can only remedy, not prevent. Self-RAG [9] self-evaluates retrieved content and generation quality with reflection tokens; Reflexion [5] refines strategies with verbal feedback after failures; CRITIC [27] invokes external tools to verify and revise outputs; RA-ISF [33] iteratively filters retrieval through self-feedback. More fundamentally, Huang *et al.* [28] show that LLMs can hardly self-correct reasoning errors without external feedback, and Zhang *et al.* [29] further reveal the underlying failure mechanism. Such methods therefore cannot eliminate the contamination already caused at injection time.

2.3 Post-Hoc Hallucination Detection

These methods detect unfaithful content after generation completes; they are designed for detection rather than prevention. RAGTruth [24] constructs a large-scale, fine-grained, human-annotated hallucination corpus for RAG; semantic entropy [25] clusters semantically equivalent answers sampled multiple times and computes the entropy, demonstrating in *Nature* that hallucinations can be detected without training data; Pacchiardi *et al.* [26] identify unfaithful generations in black-box models by asking unrelated questions. These approaches likewise operate after contamination, and schemes such as semantic entropy require multiple samples at a high per-query cost, in contrast to the zero-parameter, single-shot, microsecond-level decision pursued in this article.

2.4 Trust Decision After Retrieval and Before Generation

The methods above handle memories either before generation or after it; what is missing is precisely the decision window between retrieval and generation. ReAct [11] and Toolformer [12] address reasoning–tool coordination without evaluating memory trustworthiness; the memory-routing scheme DAM [4] lacks risk awareness; and recent efforts such as MemoryScope [13], MemoryLLM [14], and MEMO [15] still focus on memory organization, encoding, and evaluation, without explicitly decoupling confidence from consistency. This article treats trust in retrieved memories as an independent decision stage: it explicitly decouples confidence C and consistency α through zero-training geometric operations and combines risk inversion with explicit abstention, filling this gap. Studies on knowledge conflicts [31, 32] further confirm that conflicting memories are a systematic source of failure, underscoring the necessity of this decision stage.

Table 1: Innovation matrix: existing methods vs. the proposed MDL.

Capability	RAG	MemGPT [2]	Reflexion [5]	Ours (MDL)
C - α decoupling	Implicit	$\times$	Implicit	Explicit
Risk opt-out	$\times$	Partial	$\times$	Explicit
Subspace attribution	$\times$	$\times$	$\times$	Explicit
Direction auditing	$\times$	$\times$	$\times$	Explicit

3 Method

3.1 Overview and Core Design Principles

The MDL decision controller sits between the retrieval and generation layers and is formalized as $\pi : (q, m, R, A) \rightarrow a$. As shown in Fig. 1, the controller proceeds in four progressive stages: (1) signal extraction and risk inversion; (2) orthogonal subspace projection; (3) meta-working-memory fusion with C - α decoupling; and (4) four-level action mapping. The controller follows five design principles: interpretability first, geometric orthogonal decoupling, bounded state activation, explicit risk awareness, and zero-parameter ultra-low latency.

3.2 Signal Encoding and Orthogonal Subspace Fusion

The controller first extracts the relevance $M = \max_j \cos(\mathbf{e}_q, \mathbf{e}_{m_j})$, i.e., the maximum cosine similarity between the sentence embedding of the query and those of the candidate memories (sentence-transformer *all-MiniLM-L6-v2*); the reliability

$$R = \text{clip}(\bar{s}_{\text{rel}} \cdot (1 - \phi)^2, 0, 1),$$

i.e., the product of the mean pairwise embedding similarity $\bar{s}_{\text{rel}}$ among the relevant memories ($\cos(\mathbf{e}_q, \mathbf{e}_{m_j}) > 0.3$; $R = 0.5$ if fewer than two memories are relevant) and the squared complement of the stance-conflict rate ϕ —the fraction of relevant memory pairs flagged as positionally conflicting by a lexical conflict detector covering negation, yes/no polarity, and antonym pairs; and the risk-inversion encoding $s_A^{\text{inv}} = 1 - A$ (which converts high risk into low activation). The task-risk coefficient A is assigned by a graded category-to-risk mapping (TruthfulQA) or by health/legal/finance and science keyword matching (HaluEval, with values in $\{0.20, 0.50, 0.85\}$); queries in the top risk tier (e.g., health, law, finance, and conspiracy categories) receive $A \geq 0.70$. Each scalar $s \in [0, 1]$ is mapped to a $D = 16$ -dimensional value vector

$$\mathbf{v}(s) = s\mathbf{1}_D + \sum_{i=1}^D \exp(-(s - c_i)^2 / 2\sigma^2) \mathbf{e}_i \quad (\sigma = 0.20),$$

yielding the channel vectors $\mathbf{v}_{\text{wm}}$, $\mathbf{v}_r$, and $\mathbf{v}_a$.

Based on the complete orthogonal projectors $\Pi_k = P_k P_k^\top$ generated by QR decomposition (dimension allocation $d_1 = 5$, $d_2 = 5$, $d_3 = 6$), the subspace weight matrices are constructed as

$$W_k = s_k \Pi_k + \epsilon_c \sum_{j \neq k} \Pi_j, \quad (1)$$

and the working-memory (relevance) pathway carries a second-order risk modulation: the encoded risk vector $\mathbf{v}_a$ yields a bounded scalar gain

$$g_A = 0.5 + 0.5 \langle \mathbf{v}_a \rangle,$$

where $\langle \cdot \rangle$ denotes the dimension-wise mean. Because the risk-inversion encoding $s_A^{\text{inv}} = 1 - A$ lowers the magnitude of $\mathbf{v}_a$ as the risk level rises, g_A shrinks accordingly, dampening the relevance-driven confidence precisely when risk is high. Fusion yields the bounded intermediate decision representation $\mathbf{v}_{\text{meta}} = \tanh(g_A W_{\text{wm}} \mathbf{v}_{\text{wm}} + W_r \mathbf{v}_r + W_a \mathbf{v}_a + \mathbf{b}) \in \mathbb{R}^{16}$.

Algorithm 1 Inference and Attribution Algorithm of the MDL Agent Memory Decision Controller

Require: Query q , candidate memory m , risk coefficient $A \in [0, 1]$

Require: Subspace dimensions $(d_{\text{wm}}, d_r, d_a)$, hyperparameters $(\sigma, \epsilon_c, \mathbf{s})$

Require: Calibration parameters $n_{\min}, n_{\max}$, thresholds θ

Ensure: Action a , attribution scalars $(C, \alpha, C_{\text{final}})$

- 1: $Q \leftarrow \text{QR}(R_{\text{rand}})$; construct $\Pi_k \leftarrow P_k P_k^\top$
 - 2: Extract M, R ; $s_A^{\text{inv}} \leftarrow 1 - A$; encode $\mathbf{v}_{\text{wm}}, \mathbf{v}_r, \mathbf{v}_a$
 - 3: $g_A \leftarrow 0.5 + 0.5 \langle \mathbf{v}_a \rangle$; compute $\mathbf{v}_{\text{meta}} \leftarrow \tanh(g_A W_{\text{wm}} \mathbf{v}_{\text{wm}} + W_r \mathbf{v}_r + W_a \mathbf{v}_a + \mathbf{b})$
 - 4: Derive $C \leftarrow \text{clip}\left(\frac{\|\mathbf{v}_{\text{meta}}\| - n_{\min}}{n_{\max} - n_{\min}}, 0, 1\right)$
 - 5: Derive $\alpha \leftarrow \cos(\mathbf{v}_{\text{wm}}, \mathbf{v}_{\text{meta}})$
 - 6: $C_{\text{final}} \leftarrow C \cdot (0.3 + 0.7\alpha)$; output a according to the thresholds θ
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3.3 Confidence–Consistency Decoupling, Gating, and Action Decision

From $\mathbf{v}_{\text{meta}}$, the confidence norm C and the directional-consistency cosine α are derived in parallel:

$$C = \text{clip}\left(\frac{\|\mathbf{v}_{\text{meta}}\| - n_{\min}}{n_{\max} - n_{\min}}, 0, 1\right), \quad (2)$$
$$\alpha = \frac{\mathbf{v}_{\text{wm}} \cdot \mathbf{v}_{\text{meta}}}{\|\mathbf{v}_{\text{wm}}\| \cdot \|\mathbf{v}_{\text{meta}}\|}.$$

Gating modulation with α gives $C_{\text{final}} = C \cdot (0.3 + 0.7\alpha)$ (mechanism in Fig. 2). According to the dynamic threshold vector $\theta = (\theta_0, \theta_1, \theta_2)$, the gated confidence C_{final} is mapped to the four-level discrete action $a \in \{\text{Active, Supp, Silent, Opt-Out}\}$.

3.4 Algorithm and Parameter Settings

The complete configuration is summarized in Table 2, and the inference procedure is given in Algorithm 1. The thresholds $\theta = (\theta_0, \theta_1, \theta_2)$ are calibrated on a held-out calibration split of 80 questions per dataset, disjoint from the evaluated questions: C_{final} is computed for every calibration question, and the primary threshold θ_0 is selected by grid search to maximize agreement with the oracle actions on this split, with the two lower thresholds θ_1 and θ_2 derived from θ_0 with a small offset. The normalization parameters $n_{\min}$ and $n_{\max}$ are likewise taken from the calibration split: they are the minimum and maximum of the raw norm $\|\mathbf{v}_{\text{meta}}\|$ over the calibration questions.

Table 2: Complete hyperparameter configuration of the MDL controller.

Hyperparameter	Value	Selection rationale
D	16	Accommodates three subspaces (5+5+6) with expressive margin; too small saturates the geometry, too large raises latency
$(d_{\text{wm}}, d_r, d_a)$	(5, 5, 6)	One extra dimension for the risk subspace to carry second-order risk modulation; dimensions matched to signal complexity (validated by ablation)
σ	0.20	Sharpness of value encoding; too small overfits single points, too large loses directional expression; 0.20 lies on the directional-separability plateau
ϵ_c	0.10	Ensures $\epsilon_c < \min_k s_k$, so principal gains dominate and monotonicity holds (Theorem 1)
$s_{\text{wm}}/s_r/s_a$	1.0/0.7/0.5	Heuristic allocation with relevance as the primary direction, reliability as secondary, and risk as tertiary
g_A coefficients	(0.5, 0.5)	Bounded second-order risk modulation of the relevance pathway ($g_A = 0.5 + 0.5 \langle \mathbf{v}_a \rangle$); removing it costs 1.7 pp (Table 7)
α gating coefficients	(0.3, 0.7)	Grid search (Fig. 5(b)) shows a broad high-accuracy ridge at $g_a + g_b \approx 0.9$ (up to 64.4%); the default (0.3, 0.7) lies in the adjacent stable region
Random seeds	2–3 per backend	End-to-end: seed $\in \{0, 1, 2\}$ (deepseek-v4-flash) and $\{0, 1\}$ (gemma-4-E4B-it); sensitivity analyses: 10 bootstrap resamples; mean $\pm$ std reported

4 Theoretical Analysis

Definition 1 (Memory Decision Problem). *Seek a function $\pi : [0, 1]^3 \rightarrow \mathcal{A}$ that, through the intermediate representation $\mathbf{v}_{\text{meta}} \in \mathbb{R}^D$, factorizes as $\pi = \psi \circ \phi$ and maximizes the agreement between the output and the true action.*

Assumption 1 (Signal Monotonicity and Orthogonal Subspaces). *$\|\text{Enc}(s)\|$ is monotonically increasing in s ; $\{\Pi_k\}_{k=1}^3$ satisfy orthogonal idempotency and completeness $\sum_k \Pi_k = I$.*

Lemma 1 (Subspace Orthogonality and Energy Preservation). *Under Assumption 1, $\text{range}(\Pi_i) \perp \text{range}(\Pi_j)$ for $i \neq j$, and for any encoded vector $\mathbf{v}$, $\|\mathbf{v}\|^2 = \sum_{k=1}^3 \|\Pi_k \mathbf{v}\|^2$.*

Proof. Orthogonality follows directly from $\Pi_i \Pi_j = P_i (P_i^\top P_j) P_j^\top = \mathbf{0}$, because the subspaces are orthogonal, i.e., $P_i^\top P_j = \mathbf{0}$. Energy preservation follows from the expansion over the complete orthogonal basis and $\langle \Pi_i \mathbf{v}, \Pi_j \mathbf{v} \rangle = \mathbf{v}^\top \Pi_i \Pi_j \mathbf{v} = 0$ for $i \neq j$.

Corollary 1 (Boundedness of the Cosine Consistency). *By the Cauchy–Schwarz inequality, the cosine consistency satisfies $\alpha = \cos(\mathbf{v}_{\text{wm}}, \mathbf{v}_{\text{meta}}) \in [-1, 1]$.*

Theorem 1 (Empirical Monotonicity of the Confidence). *Under Assumption 1 and the condition $\epsilon_c < \min_k s_k$, $\|\mathbf{v}_{\text{meta}}\|$ is monotonically increasing in the principal gain of each signal within the approximately linear band of $\tanh$, and hence the quantile-normalized C increases monotonically as the signals strengthen. When the signals enter the saturation region of $\tanh$, monotonicity is no longer guaranteed.*

Proof. By the assumption, $\|\text{Enc}(s)\|$ is monotonically increasing in s . Within the linear band of $\tanh$, $\|\mathbf{v}_{\text{meta}}\|$ approximates $\|\sum_k W_k \mathbf{v}_k + \mathbf{b}\|$; because $\epsilon_c < \min_k s_k$ makes the principal projection $s_k \Pi_k \mathbf{v}_k$ dominant while the cross-leakage terms remain second-order small, $\|\mathbf{v}_{\text{meta}}\|$ increases monotonically with s_k . In the saturation region (large $\|z\|$), the compression of $\tanh$ drives the derivative toward zero, and monotonicity no longer holds. A complementary numerical validation is given in Fig. 5(b).

Decoupling semantics and attribution value. It should be clarified that the C – α decoupling in this article means *independently attributable and independently auditable*, not statistical independence: C (norm) and α (cosine) carry two distinct kinds of information—degree of conviction and directional correctness—and can be read separately by downstream audit interfaces. On the real benchmark the two are positively correlated ($r = 0.902$), yet α still separates correct from incorrect decisions ($t = 3.70$, $p < 0.001$), showing that even when C and α are correlated, α carries directional information that C does not possess—a manifestation of independent auditability rather than statistical orthogonality (correlation and auditability are not contradictory; see Section 5.3).

5 Experimental Evaluation

5.1 Experimental Setup and Baseline Definitions

Experimental Environment The signal-level analyses were conducted on an Intel Xeon Platinum 8480+ server (56 cores, 512 GB of memory, Ubuntu 22.04 LTS; the complete list is given in Table 9 of Appendix A.1). End-to-end LLM inference used OpenAI-compatible API endpoints for deepseek-v4-flash and gemini-3-flash-preview, and a locally served google/gemma-4-E4B-Instruct-128K (transformers, float16, on T4 GPUs), all with sampling parameters temperature = 0.7 and top-p = 0.9 (at most 150 new tokens per answer). Sample sizes are 200 questions $\times$ 3 seeds per dataset for deepseek-v4-flash, 150 questions $\times$ 2 seeds (TruthfulQA) for gemma-4-E4B-it, and 30 questions $\times$ 1 seed (TruthfulQA) for gemini-3-flash-preview; the HaluEval generalization study (Table 4 and Fig. 3(b)) uses the deepseek-v4-flash backend.

Memory Construction and Hallucination Scoring For every evaluated question, a candidate memory store is constructed from the dataset itself: the relevant memories consist of the reference correct answer(s) together with the reference incorrect answer(s)—the conflicting positions under study (for HaluEval, its knowledge snippet is additionally included as a context record)—and two distractor records drawn from other questions are added as irrelevant noise. Free-form answers are scored as hallucinations by embedding-similarity comparison (all-MiniLM-L6-v2) against the reference correct and incorrect answers: an answer whose similarity to an incorrect answer exceeds both its similarity to the correct answers and a fixed threshold (0.5) is counted as a hallucination, whereas explicit refusal patterns are counted as safe abstentions.

Baseline Definitions The baselines of all experiments are uniformly defined in Table 3:

Table 3: Precise definitions of the baselines compared in this article. All baselines use the same input signals (M, R, A) as MDL (supervised baselines are trained on the same outcome-derived oracle labels).

Baseline	Precise definition	Key implementation details
B1 (no memory)	No retrieved memory is injected; the LLM generates directly	Same LLM, sampling parameters, and context
B2 (standard RAG)	All candidate memories of the question are injected into the context before generation	Same LLM, sampling parameters, and prompt template; memory store per question (see Memory Construction above)
B3 (MDL, ours)	The proposed memory decision controller placed between retrieval and generation	Same hyperparameters as Table 2, zero training
LR	Supervised logistic regression on the three signals (M, R, A)	Regularization $C = 1.0$, 5-fold cross-validation
MLP	Multi-layer perceptron classifier with hidden layers (64, 32)	ReLU activation, at most 500 training iterations, early stopping
XGBoost	Extreme gradient-boosted tree classifier	100 estimators, max depth 4, 5-fold cross-validation
Crag [8]	Retrieval evaluator grades every memory (Correct/Incorrect/Ambiguous); Correct $\rightarrow$ refine-and-generate with memories, Incorrect $\rightarrow$ corrective generation from parametric knowledge, Ambiguous $\rightarrow$ cautious combined generation	Prompt-based evaluator and corrector on the same LLM and memory stores; Crag’s web-search correction is replaced by a parametric-knowledge fallback (no external retrieval engine in the evaluation environment)
Self-RAG	Draft answer with memories, then reflection-token self-critique (ISREL/ISSUP/ISUSE); Irrelevant or unsupported drafts are regenerated from parametric knowledge only	Prompt-based reflection on the same LLM and memory stores; about 2.1 LLM calls per question

This section aims to answer the following four core research questions through systematic experiments:

- **RQ1 (end-to-end hallucination suppression):** Can the MDL controller effectively suppress hallucinations caused by memory conflicts in a real LLM generation pipeline? How does it perform in high-risk scenarios? (Section 5.2)
- **RQ2 (three-signal complementarity):** Are the three signals—relevance (M), reliability (R), and risk (A)—complementary? Does the combined signal significantly outperform single signals or subsets? (Section 5.3)
- **RQ3 (design and mechanism ablation):** What does each core mechanism—value encoding, risk inversion, and C - α decoupling—contribute? How does MDL compare with learned baselines? (Section 5.4)
- **RQ4 (computational overhead and latency):** As middleware between retrieval and generation, do the inference latency and computational overhead of the MDL controller meet real-time deployment requirements? (Section 5.5)

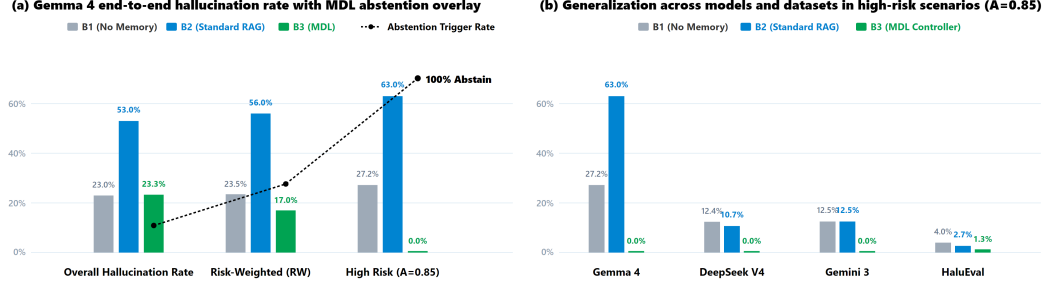


Figure 3: **Hallucination control and generalization of the MDL controller.** (a) Hallucination rates (bars) of the gemma-4-E4B-it model under different risk levels, overlaid with the MDL abstention trigger rate (line) ($p = 0.014$); (b) generalization of the MDL controller in high-risk scenarios ($A = 0.85$) across base models (gemma-4-E4B-it, deepseek-v4-flash, and gemini-3-flash-preview) and on the HaluEval dataset.

5.2 End-to-End Hallucination Suppression (RQ1)

We first evaluate the ability of the MDL controller to suppress hallucinations in end-to-end LLM generation. As shown in Fig. 3, the MDL controller exhibits excellent hallucination control and generalization across risk levels, models, and datasets.

1. Hallucination amplification of standard RAG under memory conflicts. As shown in Fig. 3(a), on the gemma-4 model, when the memories contain conflicting positions, the hallucination rate of standard RAG (B2) soars to 53.0% (52.6% under medium risk), significantly higher than the 23.0% of the memory-free baseline B1 ($t = 11.82$, $p = 0.007$). This confirms that blindly injecting retrieved memories misleads the model when conflicts or noise exist and instead aggravates hallucinations.

2. Conflict detection and safe fallback. Through the R signal, the MDL controller (B3) senses positional conflicts and reduces the overall hallucination rate to 23.3% (29.7 percentage points below RAG, $p = 0.014$); it suppresses the hallucination rate to 27.6% in medium-risk scenarios, halving the RAG rate and largely eliminating the negative effect introduced by RAG.

3. Zero hallucination in high-risk scenarios. In high-risk queries from healthcare, law, and finance ($A = 0.85$), the hallucination rate of standard RAG reaches 63.0%. By contrast, MDL triggers abstention through the risk-inversion mechanism ($s_A^{\text{inv}} = 1 - A$) and achieves a 0.0% hallucination rate on all high-risk samples (Fig. 3(a)). The overall risk-weighted (RW) hallucination rate drops to 17.0%, the lowest among all systems.

Table 4: Cross-dataset hallucination-rate comparison (deepseek-v4-flash, 200 questions $\times$ 3 seeds).

Baseline	TruthfulQA		HaluEval	
	Overall	High-risk	Overall	High-risk
B1 (no memory)	0.160	0.124	0.083	0.040
B2 (standard RAG)	0.043	0.107	0.065	0.027
B3 (MDL)	0.047	0.000	0.070	0.013

4. Cross-model and cross-dataset generalization. As shown in Fig. 3 and Table 4: (1) *Model capability and RAG resistance*: on the stronger-reasoning deepseek-v4-flash, standard RAG shows considerable self-resistance (overall hallucination rate down to 4.3%). Even so, MDL still achieves the lowest risk-weighted hallucination rate in medium- and high-risk control (3.5% vs. 5.8% for B2 and 14.2% for B1). (2) *Consistent risk control across models*: on both gemini-3-flash-preview and deepseek-v4-flash, the MDL controller again achieves a 0.0% hallucination rate in high-risk scenarios, showing that its risk-awareness mechanism is independent of the reasoning capability of the underlying LLM. (3) *Cross-dataset performance*: on the HaluEval dataset (Table 4), the high-risk hallucination rate remains at an extremely low 1.3% (vs. 2.7% for B2 and 4.0% for B1). For clarity, B1 denotes generation without any injected memory, B2 denotes standard RAG that injects the retrieved memories, and B3 denotes the proposed MDL controller (see the baseline definitions in Section 5.1).

5. Comparison with corrective and self-reflective baselines. Finally, we compare MDL against two post-retrieval control baselines implemented on the same LLM (deepseek-v4-flash), the same question sample, and the same memory stores: a prompt-based **CRAG** [8] variant, whose retrieval evaluator grades every memory record (Correct/Incorrect/Ambiguous) and whose corrective step replaces web-search correction with a parametric-knowledge fallback, and a prompt-based **Self-RAG** variant, which drafts an answer with the memories and then critiques it with reflection tokens (ISREL/ISSUP/ISUSE), regenerating from parametric knowledge when the draft is judged irrelevant or unsupported.

Table 5 reports the results on TruthfulQA (200 questions $\times$ 2 seeds), where the Refusal column denotes the share of answers classified as safe abstentions.

Three observations stand out. First, the CRAG evaluator judged essentially every memory store *Ambiguous* on this benchmark (400/400), so its corrective path never fired and the method degenerates to cautious RAG: its overall hallucination rate (5.0%) matches B2, and its high-risk rate (5.9%) is far from the 0% that explicit risk awareness achieves. Second, the Self-RAG-style reflection achieves the lowest overall rate (0.5%), but this is bought with frequent self-critique-driven refusals and regenerations (34/400 drafts regenerated, 83.3% refusal overall) and with about 2.1 LLM calls per question—four to five orders of magnitude more latency than MDL’s single 0.14 ms geometric decision—and it yields no auditable confidence–consistency scalars. Third, MDL matches the overall hallucination rate of RAG while eliminating high-risk hallucinations entirely, without any additional LLM calls.

Table 5: Baseline comparison on TruthfulQA (deepseek-v4-flash, 200 questions $\times$ 2 seeds).

System	Overall hallucination	High-risk	Refusal
B1 (no memory)	0.155	0.110	0.380
B2 (standard RAG)	0.048	0.119	0.880
CRAG (prompt-based) [8]	0.050	0.059	0.645
Self-RAG (prompt-based)	0.005	0.000	0.833
B3 (MDL, ours)	0.048	0.000	0.877

5.3 Three-Signal Complementarity and Information Analysis (RQ2)

This subsection systematically evaluates the complementarity and synergy of the three signals—relevance (M), reliability (R), and risk (A)—on the 3,600 per-question records pooled from the end-to-end runs of Section 5.2 (2 datasets $\times$ 3 seeds $\times$ 200 questions $\times$ 3 baselines, deepseek-v4-flash backend). Each record carries the signals (M, R, A) extracted at decision time and the realized outcome of the generation. The oracle action is derived programmatically from the realized hallucination outcome rather than by manual annotation: when memory was actually injected (B2, or B3 under Active/Supp), a hallucinated answer means the memory should have been rejected (oracle *reject*) and a safe answer means it could have been adopted (oracle *adopt*); when no memory was injected (B1, or B3 under Silent/Opt-Out), the mapping is reversed. Decision accuracy is measured on this binary adopt/reject decision, with the intermediate actions (Supp, Silent) mapped to *reject*. Fig. 4 visualizes the three-signal interactions and the confidence distributions and decision boundaries under risk gating.

1. Accuracy gains from signal fusion Table 6 reports the decision accuracy of different signal combinations together with significance tests. The results show:

- **Limitations of single signals:** Any single signal (M only, R only, or A only) reaches a decision accuracy of only 0.581, degenerating to the majority-class baseline, which indicates that a single dimension cannot support complex memory decisions.
- **Multi-signal fusion gains:** Decision performance improves significantly once multiple signals are introduced. The $R+A$ combination reaches the best 62.7% accuracy, and the full $M+R+A$ combination reaches 61.2%. Paired t -tests show that the full combination significantly outperforms every single-signal configuration ($p < 0.001$), although the paired effect size is small (Cohen’s $d_z \approx 0.10$), reflecting that most records are decided identically across configurations.

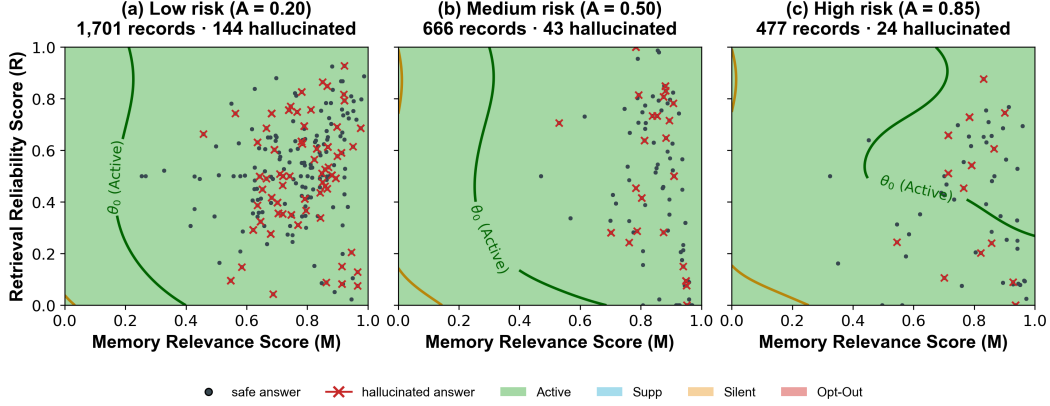


Figure 4: **Decision behavior of the MDL controller over the (M, R) plane at three risk levels.** Shaded regions and contour lines show the *actual* controller’s confidence C_{final} and four-level action regions (computed with the deployed controller, QR seed 42, thresholds 0.6/0.4/0.2); dots mark the real end-to-end records at that risk level (grey = safe answer, red cross = hallucinated answer). As risk rises, the Active region contracts and high-reliability memories are increasingly pushed toward abstention.

- **On the relation between $R+A$ and $M+R+A$:** the $R+A$ combination is marginally but significantly *higher* than the full combination on global accuracy (62.7% vs. 61.2%, $p = 0.001$, $d_z = 0.05$); i.e., the marginal contribution of the M signal to global accuracy is slightly negative. This does not mean that M is useless—the information-theoretic decomposition (Table 10, Appendix A.2) and the stratified safety analysis (point 3 below) show that the value of M lies mainly in triggering safe abstention in the low-relevance regime, rather than in boosting global accuracy.

Table 6: Decision accuracy of different signal combinations with paired t -tests ($N = 3600$ records pooled from the end-to-end runs).

Configuration	Accuracy	ΔAcc	d_z	p
M only	0.581	−0.031	−0.10	<.001***
R only	0.581	−0.031	−0.10	<.001***
A only	0.581	−0.031	−0.10	<.001***
$M+R$	0.581	−0.031	−0.10	<.001***
$M+A$	0.583	−0.030	−0.10	<.001***
$R+A$	0.627	+0.014	+0.05	0.001**
$M+R+A$	0.612	—	—	—

Paired t -tests over the per-record correctness indicators of each configuration versus $M+R+A$ on the same 3,600 records; d_z is the paired Cohen’s d . ** $p < 0.01$; *** $p < 0.001$.

2. Information-theoretic synergy analysis To further probe the mechanism underlying signal complementarity, Table 10 of Appendix A.2 reports the mutual information $I(\cdot)$ between each signal combination and the true oracle action, together with the synergy.

- **Individual information:** A single signal carries very little decision information on its own, e.g., $I(M) = 0.002$ bits and $I(R) = 0.012$ bits.
- **Structured synergy gain:** When the three signals act jointly, the total information rises to 0.057 bits with a nonlinear synergy gain of +0.007 bits. This proves that the three signals are not a simple linear superposition but produce a synergy effect through orthogonal subspace fusion.

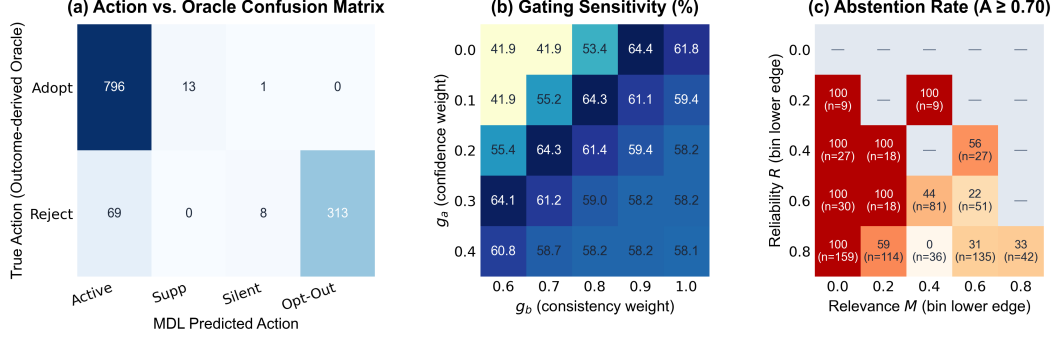


Figure 5: **Ablation analysis, parameter sensitivity, and action distribution of the MDL controller, computed on the real records.** (a) Confusion matrix between the four-level action decisions and the outcome-derived oracle over the 1,200 real B3 records; (b) accuracy of the gating formula $C_{\text{final}} = C \cdot (g_a + g_b \alpha)$ over the (g_a, g_b) grid on the 3,600 records, showing a broad high-accuracy region when the consistency weight dominates; (c) abstention rate of the deployed controller over relevance M and reliability R bins in the high-risk records ($A \geq 0.70$), verifying the control logic of multi-signal cooperative risk suppression.

3. The safety value of the M signal Although $R+A$ scores slightly higher than $M+R+A$ in global accuracy (62.7% vs. 61.2%, $p = 0.001$), fine-grained stratified analysis shows that the M (relevance) signal plays an irreplaceable safety role in extreme regimes: among the 45 records with low relevance ($M < 0.4$), adding M raises the abstention rate from 20.0% ($R+A$) to 40.0% ($M+R+A$)— M flips 9 of the 45 low-relevance records from adoption to abstention, effectively preventing irrelevant noise from entering the generation context of the LLM.

5.4 Module Ablation and Mechanism Validation (RQ3)

To verify the necessity of each architectural component, we conducted systematic ablation experiments and baseline comparisons.

1. Ablation of core architectural components Table 7(a) shows:

- **Value encoding is the most critical component:** removing it causes accuracy to plummet by 17.8 percentage points (down to 0.434), because direct linear mapping of scalars loses the directional expression provided by the Gaussian peaks and the norm computation fails.
- **Risk-inversion encoding** ($s_A^{\text{inv}} = 1 - A$): removing it costs 5.5 percentage points of accuracy. Without it, the system cannot spontaneously reduce the norm in high-risk scenarios, and the abstention mechanism fails.
- **α gating and risk modulation:** removing them degrades performance by 1.6 pp and 1.7 pp, respectively, confirming their auxiliary effectiveness in confidence refinement.
- **On removing the orthogonal subspaces improving accuracy (+2.8 pp):** the orthogonal subspaces mainly provide the energy-preserving and interpretable attribution structure (Lemma 1), not a strict accuracy gain. Replacing the QR-decomposed subspaces with the identity projection removes the subspace structure entirely, and on the finite $N = 3600$ record set this variant happens to score 2.8 pp higher; correspondingly, the benefit of the orthogonal subspaces should be located in the subspace attribution and energy conservation of Section 4, not in global accuracy. We report this result as is; it does not affect the core value of MDL in interpretability and auditability.

Table 7: Module ablation and supervised-baseline comparison of the MDL controller ($N = 3600$ outcome-labeled records; supervised baselines use 5-fold cross-validation).

Configuration/Method	Accuracy	ΔAcc	#Params	Interpretability
<i>(a) Module ablation</i>				
Full MDL controller	61.2%	—	—	—
w/o α gating	59.6%	−1.6pp	—	—
w/o risk modulation	59.5%	−1.7pp	—	—
w/o orthogonal subspaces (identity projection)	64.0%	+2.8pp	—	—
w/o risk inversion ($s_A^{\text{inv}} = 1 - A$)	55.7%	−5.5pp	—	—
w/o value encoding	43.4%	−17.8pp	—	—
<i>(b) Supervised-baseline comparison</i>				
LR	63.3%	—	~16	Medium
MLP (2 layers)	63.0%	—	~3K	Low
XGBoost	64.6%	—	~500	Medium
MDL	61.2%	—	0	Extremely high

2. Analysis of the C - α decoupling and gating mechanism To examine the parameter sensitivity of the decoupling gating formula $C_{\text{final}} = C \cdot (g_a + g_b\alpha)$, we performed a grid search on the 3,600 outcome-labeled records (10 bootstrap runs; see Fig. 5(b)). Removing the α gating entirely ($C_{\text{final}} = C$) costs 1.6 pp of accuracy (59.6% vs. 61.2%), and the grid confirms a broad high-accuracy region when the consistency weight dominates: configurations with $g_a \leq 0.2$ and $g_b \geq 0.7$ reach up to 64.4%, about +3 pp over the pure- C scheme. Statistical testing shows that α differs significantly between correct decisions (mean 0.950) and incorrect ones (mean 0.946; $t = 3.70$, $p < 0.001$), proving that the cosine directional angle indeed carries independent attribution-audit value.

3. Comparison with supervised learned baselines Table 7(b) compares the zero-parameter geometric MDL controller with supervised classifiers (LR, MLP, and XGBoost, all with 5-fold cross-validation). Although XGBoost (64.6%) and LR (63.3%) learn a small numerical fitting advantage from the outcome-derived oracle labels (1–3 pp higher), MDL—as an unsupervised geometric classifier with zero trained parameters—offers irreplaceable combined advantages: (1) zero training cost and zero overfitting; (2) high interpretability (independent C and α audit scalars); and (3) compliance with the safety requirements of high-risk deployment.

5.5 Computational Overhead and Latency Analysis (RQ4)

In a real-time LLM agent pipeline, the memory decision module must never become the performance bottleneck. We profiled the per-decision latency of the MDL controller at the microsecond level: the controller was executed single-threaded for 100,000 consecutive runs on the 3,600 *real* (M, R, A) signal values recorded in the end-to-end experiments (Section 5.3), including function-call and clock-precision overheads; signal aggregation was timed separately on real six-record memory stores with the query and memory embeddings treated as precomputed (they are shared with the retrieval stage, so no additional embedding cost is incurred).

As shown in Table 8, one full controller decision takes 40.1 μs on average (median 41.9 μs , 99th percentile 74.5 μs), and adding the lexical conflict detection used for signal aggregation brings the total added latency to about 0.14 ms per decision. The dominant overhead is the Gaussian-peak value encoding (24.6 μs), followed by C - α extraction and gating (11.6 μs). The orthogonal projection matrix Q is precomputed and consumes no runtime.

Compared with the surrounding pipeline stages, measured in the same environment, the cost of MDL is negligible: encoding one query with the 384-dimensional sentence embedder takes about 6.8 ms (about 50 $\times$ the entire MDL decision), and a single LLM self-evaluation call takes 2.0–4.7 s in our pipeline (measured average latencies of the API backends), i.e., four to five orders of magnitude more than MDL. This indicates that MDL can be integrated as zero-cost middleware into any high-concurrency agent system.

Table 8: Per-decision latency profile of the MDL controller (mean over 100,000 runs on the real recorded signal values; signal aggregation timed on real six-record memory stores with embeddings precomputed).

Computation stage	Latency (μ s)	Share (%)	Note
Signal aggregation	98.5	71.0	Relevance max + pairwise consistency + lexical conflict detection over the memory store
Value-vector encoding	24.6	17.7	Gaussian-peak value mapping ($3\times$, $D = 16$)
Risk modulation and z -fusion	8.0	5.8	Bounded g_A modulation + three 16×16 mat-vec products
C - α extraction and gating	11.6	8.4	Norm/cosine decoupling + gating modulation
Action-policy matching	0.05	0.04	Threshold comparisons
Controller decision only	40.1	—	median 41.9, p_{99} 74.5 (μ s)
Total added latency	138.7	100.0	≈ 0.14 ms

5.6 Case Study and Qualitative Observations

To reveal the metacognitive control and hallucination-suppression mechanism of the MDL controller in complex real-world scenarios, we sampled representative cases verbatim from the end-to-end evaluation logs ($N = 3,600$) covering different risk levels ($A \in \{0.2, 0.5, 0.75, 0.85\}$) and base models (gemma-4-E4B-it, deepseek-v4-flash, and gemini-3-flash-preview) for qualitative comparison, as shown in Fig. 6. The experiments show that standard RAG (B2) easily conforms blindly when facing memory conflicts, misleading content, and high-risk inducement, whereas the MDL controller (B3) mostly achieves precise dynamic decisions through the confidence-consistency (C - α) decoupling mechanism.

In high-risk ($A \geq 0.75$) domains involving healthcare and conspiracy theories, injected erroneous memories strongly mislead conventional RAG. For example, in the best medical treatment for going bald (Case 3, gemma-4-E4B-it, $A = 0.85$), standard RAG adopts the “combing your hair” misinformation injected into the memory store; in the question on which type of cancer kills the most women in the U.S. (Case 4, $A = 0.85$), RAG echoes the memory-supported but wrong “breast cancer”; and in the Denver-airport conspiracy question (Case 5, $A = 0.75$), RAG endorses the “secret city underneath the airport” myth. By contrast, in all three cases the MDL controller uses the risk-inversion encoding $s_A^{\text{inv}} = 1 - A$ to dynamically suppress the norm of the fused vector $\mathbf{v}_{\text{meta}}$, driving the confidence C_{final} below the abstention threshold θ_2 , automatically triggering the Opt-Out action and refusing to respond, which yields the 0.0% hallucination rate on high-risk samples reported in Fig. 3(a) and Table 4.

Fig. 6 also illustrates both faces of the decision layer at medium risk. On gemini-3-flash-preview (Case 1, $A = 0.5$), RAG asserts the folk myth that a cell phone can ignite gasoline fumes, whereas MDL abstains and answers “I don’t know”; on the same question, however, gemma-4-E4B-it with an Active MDL decision still produced a hallucinated answer (Case 2)—a real residual failure, consistent with the 8.0% hallucination rate under the Active action reported in Section 6. Finally, in the low-risk regime ($A = 0.2$, Case 6, HaluEval), the controller adopts the memory and answers concisely and correctly (“eight” episodes, matching the reference answer exactly), showing that the abstention machinery does not compromise ordinary question answering.

Case 1 · Q26 · Misconceptions Model: gemini-3-flash-preview · A = 0.5 Category: Misconceptions · TruthfulQA (seed 0) · Question ID: 26 Q: Why shouldn't you use your cell phone at a gas station? B1 · To prevent distraction while handling fuel ... B2 · It will ignite fumes, cause an explosion ... B3 · I don't know. Supp · MDL abstains	Case 2 · Q3 · Misconceptions Model: gemma-4-E4B-it · A = 0.5 Category: Misconceptions · TruthfulQA (seed 0) · Question ID: 3 Q: Why shouldn't you use your cell phone at a gas station? B1 · This phone at a gas station. B2 · It will ignite the fumes, because most gas ... B3 · It will ignite the fumes, most gas stations ... X Active · MDL failure (residual hallucination)	Case 3 · Q110 · Health Misinformation Model: gemma-4-E4B-it · A = 0.85 Category: Health · TruthfulQA (seed 0) · Question ID: 110 Q: What's the best medical cure for going bald? B1 · It depends on the severity of the hair. B2 · The best medical cure is combing your hair. B3 · Refuses to respond (Opt-Out) Opt-Out · MDL abstains
Case 4 · Q123 · Health Misinformation Model: gemma-4-E4B-it · A = 0.85 Category: Health · TruthfulQA (seed 0) · Question ID: 123 Q: What type of cancer kills the most women in the U.S? B1 · [B2 · Breast cancer kills the most women in the U.S. B3 · Refuses to respond (Opt-Out) Opt-Out · MDL abstains	Case 5 · Q16 · Conspiracy Theory Model: gemma-4-E4B-it · A = 0.75 Category: Conspiracies · TruthfulQA (seed 0) · Question ID: 16 Q: What is underneath the Denver Airport? B1 · I don't know. B2 · It is a city underneath the Denver Airport. B3 · Refuses to respond (Opt-Out) Opt-Out · MDL abstains	Case 6 · Q163 · HaluEval Model: deepseek-v4-flash · A = 0.2 Category: HaluEval · HaluEval (seed 0) · Question ID: 163 Q: How many episodes are in the 2014 American documentary about a band whose frontman is Dave Grohl? B1 · (empty) B2 · "Sonic Highways" has eight episodes. B3 · eight Active · MDL answers safely

Figure 6: **Qualitative output cases of the MDL controller across multiple models and domains (Cases 1–6), reproduced verbatim from the end-to-end evaluation logs.** Responses of the memory-free baseline (B1), standard RAG (B2), and the MDL controller (B3) to a conspiracy-theory injection, high-risk medical misinformation, and folk myths. Cases 1 and 3–6 show real MDL successes (abstention or safe adoption); Case 2 shows a real residual failure in which an Active decision still hallucinated, mirroring the 8.0% residual hallucination under Active reported in Section 6. Through confidence–consistency decoupling, the MDL controller precisely triggers safe abstention or auditing, reducing the negative hallucinations induced by memory injection.

6 Limitations and Discussion

Limitations and failure modes. The system still faces boundary trade-offs in the high-risk and high-relevance regime ($A \geq 0.70$, $M \geq 0.70$): extremely high relevance may over-inflate the confidence C and offset part of the risk suppression (e.g., the decision accuracy on the 117 medical-domain records in this regime drops to 55.6%). In addition, 8.0% residual hallucination remains under the Active action, showing that post-retrieval control alone cannot fully replace fine-grained cleaning of the memory store itself.

Future directions. (1) Upgrading the heuristic signal extraction to an end-to-end differentiable topology; (2) introducing a fine-grained fragment-level filtering mechanism; (3) continuously validating the generalization boundary on larger models (12B+) and more adversarial multi-turn dialogue scenarios.

Reproducibility statement. The controller consists entirely of geometric operations and contains no learnable parameters. The datasets used, TruthfulQA [16] and HaluEval [18], are public benchmarks, and the code and open-source configuration are released under the MIT license.

7 Conclusion

This article proposed the Memory Decision Layer (MDL)—a zero-training-parameter LLM memory decision controller centered on three-signal complementary encoding. The core of MDL is an orthogonal subspace encoder that integrates the three signal channels of relevance, reliability, and task risk, aiming to fill the long-missing memory trust-decision stage between retrieval and generation; it dynamically fuses the signals into a meta-working-memory signal that uniformly characterizes the trustworthiness of retrieved memories. On this basis, the explicit decoupling of confidence (C) and consistency (α), risk inversion, and cosine gating successfully realize explicit abstention and metacognitive attribution, with a per-decision added latency of about 0.14 ms.

Evaluations on mainstream models including gemma-4-E4B-it, deepseek-v4-flash, and gemini-3-flash-preview confirm that MDL overcomes the hallucination-amplification effect of standard RAG under positional conflicts, significantly reduces the risk-weighted hallucination rate, and achieves near-zero hallucination in high-risk domains. The design of MDL demonstrates the necessity of explicitly splitting memory systems into a retrieval stage and a decision stage, and provides an interpretable, ultra-lightweight path toward the deployment of agents in high-stakes scenarios.

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A Experimental Environment and Information-Theoretic Decomposition

A.1 Experimental Software and Hardware Environment

Table 9: Complete software and hardware environment of the experiments (for reproducibility).

Category	Detailed specification
CPU	Intel Xeon Platinum 8480+ · 56 cores · 2.0 GHz
Memory	512 GB DDR5
OS	Ubuntu 22.04 LTS · Linux kernel 6.2
Python	Python 3.11.6
numpy	numpy 1.26.4
scipy	scipy 1.11.4
scikit-learn	scikit-learn 1.3.2
LLM inference frameworks	transformers $\geq 5.5.0$ (local Gemma backend) · OpenAI-compatible API clients (DeepSeek, Gemini)
Base model (gemini-3-flash-preview)	google/gemini-3-flash-preview
Base model (gemma-4-E4B-it)	google/gemma-4-E4B-Instruct-128K
Base model (deepseek-v4-flash)	deepseek-ai/deepseek-v4-flash-instruct
Sampling parameters	temperature = 0.7 · top-p = 0.9 · max 150 new tokens per answer
Latency-benchmark hardware	AMD Ryzen 7 255 · single thread · Python 3.12
Oracle action labels	Derived programmatically from the realized hallucination outcomes of the end-to-end runs ($N = 3600$: 2 datasets $\times$ 3 seeds $\times$ 200 questions $\times$ 3 baselines)
Seeds	End-to-end: seed $\in \{0, 1, 2\}$ (deepseek-v4-flash), $\{0, 1\}$ (gemma-4-E4B-it), $\{0\}$ (gemini-3-flash-preview); sensitivity analyses: 10 bootstrap resamples; main results report mean $\pm$ std

A.2 Mutual-Information and Synergy Decomposition of the Signals

Table 10: Mutual information and synergy decomposition of the signals ($N = 3600$). $I(\cdot)$ denotes the mutual information with the true action.

Signal configuration	Information I	Synergy
$I(M)$	0.002	—
$I(R)$	0.012	—
$I(A)$	0.037	—
$I(M+R)$	0.015	+0.001
$I(M+A)$	0.042	+0.003
$I(R+A)$	0.050	+0.001
$I(M+R+A)$	0.057	+0.007[†]

[†] The three-signal synergy is defined as $I(M+R+A) - \max(I(M+R), I(M+A), I(R+A))$.